Akshat Srivastava

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Skills

- **Programming Languages:** Embedded C, C, Python, Assembly, Java
- **Microcontrollers & Boards:** ESP32, Arduino, Raspberry Pi
- **Embedded & IoT:** Sensor Interfacing, Real-Time Monitoring, IoT Architecture
- **Communication Protocols:** UART, SPI, I2C, GSM, GPS, nRF24L01
- **EDA & Simulation:** MATLAB, Simulink, Verilog, Keil µVision
- **Design Tools:** AutoCAD, Creo, KiCad
- **Soft Skills:** Problem Solving, Communication Skills, Attention to Detail, Adaptability

Internship

Super Tannery Ltd. – Observational Intern Jun 2025 – Jul 2025

- Analyzed end-to-end manufacturing workflows across 5+ production stages in a large-scale shoe manufacturing unit.
- Monitored production efficiency for a line producing 1,000+ units per day.
- Supported quality control inspections, helping identify defects and reduce rework during daily checks.
- Observed industrial automation and process optimization practices used in high-volume manufacturing.

Projects

Smart Energy Meter (IoT-Based System) Dec 2023

- Engineered an IoT-enabled smart energy meter using ESP32 and PZEM-004T for real-time power monitoring.
- Integrated continuous voltage, current, power, and energy with continuous sampling at 2–3 second intervals.
- Developed a web dashboard to visualize live electrical parameters for 24/7 monitoring.
- Enabled threshold-based alerts to notify users when energy usage exceeds predefined limits.

Technologies: ESP32, Embedded C, Sensors, WiFi, IoT, Web Server

Wireless Transmitter & Receiver for Actuator Control Mar 2025

- Architected a wireless actuator control system using Arduino Nano and nRF24L01.
- Deployed reliable wireless communication over long-range distances in interference-prone environments.
- Reduced system cost by over 60%, lowering unit cost from ₹5,000+ to under ₹2,000.
- Optimized hardware debugging and protocol tuning to ensure consistent actuator response.

Technologies: Arduino Nano, nRF24L01, Embedded C, Wireless Communication

Autonomous IoT Boat for Waterborne Disease Detection (SIH 2025) Sep 2025

- Designed an autonomous IoT-based embedded system using ESP32 and water-quality biosensors.
- Integrated GPS and ultrasonic sensors to enable obstacle detection and autonomous navigation.
- Streamed real-time sensor data via GSM to a centralized web portal during flood scenarios.
- Improved operational safety by enabling remote monitoring without human intervention.
  Technologies: ESP32, IoT, GPS, GSM

Certificates

- TCS iON Career Edge – Young Professional Aug-2025
- AI for OCI and Cloud Applications – Oracle Jan - 2025
- Embedded C Programming for Microcontrollers – Udemy Aug -2024
- Python Programming – Scaler Dec – 2023

ACHIEVEMENTS

- Secured 1st position in Circuit Solving at Tech Marathon (SEEE). Dec - 2023
- Secured 2nd position in Robo Soccer at Xpecto, IIT Mandi Mar-2025
- Secured 1st position in Tech Fest at Pushp Gujaral Science City, Jalandhar Sep - 2024
- Secured 3rd position in Robo Sumo at Cognitia Feb - 2025
- Secured 2nd position in Robo Offroad Racing at Utkansh, NIT Jalandhar April - 2025

EDUCATION

Bachelor of Technology (B.Tech) – Electronics and Communication Engineering (ECE)
Lovely Professional University, Punjab
CGPA: 6.84 2023 – Present

Intermediate (Class XII) – 75%
K.R. Education Centre, Kanpur 2022 – 2023

Matriculation (Class X) – 90%
K.R. Education Centre, Kanpur 2020 – 2021